

and does so with high precision. The biggest advantage of this method is that it is inexpensive and very accurate, and perfect for creating multiple identical replicas in a manufacturing process. The biggest challenge is of course perfecting the material used as a mold or stamper, and of course creating the complex structure on that material, as a negative – when you want the resist layer to have a peak then the mold has to have a valley there, and vice versa.

A method called thermoplastic nanoimprinting uses the technique of heating up the resist layer to soften it so as to receive the imprint from the

directly. There are a few more top down methods that are used, but they're not as common as the three we have detailed here.

Bottom up

Think about building something with Lego. You have a whole box of pieces, and you assemble them into whatever structure you envisage. This is the simple logic of bottom up manufacturing, and it's not limited to the nanoscale.

Almost everything we use in our daily lives comprises of parts that were assembled, because this results in the ability to add complexity into

with a bottom up approach. For one thing, it's a lot cheaper to do it this way, as there is less wastage and the machines needed to do bottom up manufacturing are simpler and cheaper to operate.

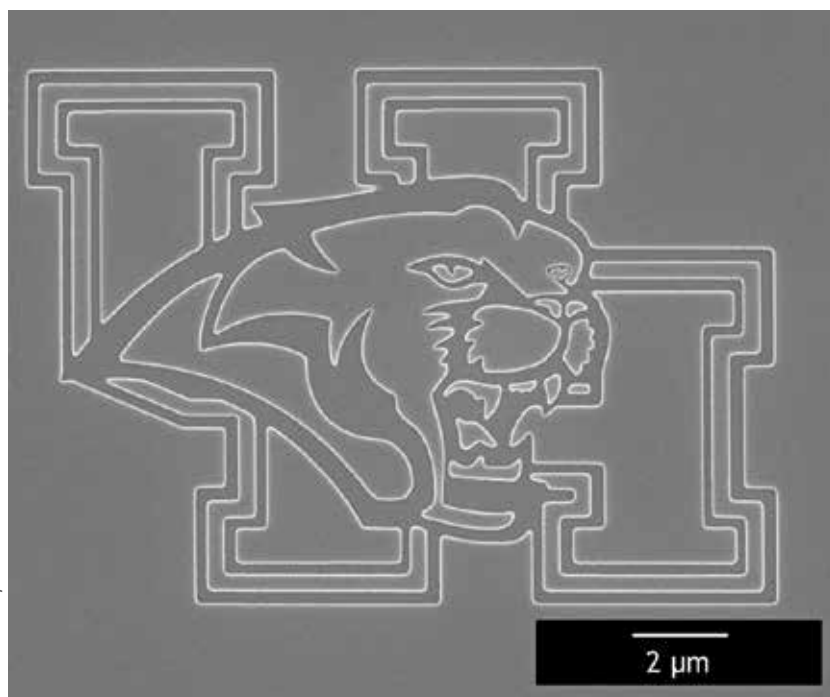
Nature is a great teacher, and we've only just started to understand her teachings and heed her advice. Life itself is a bottom up nano-process, as are almost all organic activities.

There are various methods used to achieve such a bottom up manufacturing of nanomaterials, but obviously it's an additive process. Sometimes a gas is used to make it condense as a liquid (a few atoms at a time) on to the substrate – this method is called chemical vapour deposition, and it has several dozen sub-methods of building nanomaterials.

We are even using nature itself to help us manufacture many nanoparticles and nanomaterials, a common method being researched is to use DNA to help assemble complex molecules, much like it does already anyway. Of course we'd have to figure out how to program DNA to viably produce loads of the molecules and nanomaterials we need, which is easier said than done.

Self assembly is also a part of bottom up nanomanufacturing, except that the process is automated using nature. Look at it this way, we don't have a good explanation as to why self assembly happens in nature – molecules assembling themselves into set shapes. But we may not need to understand why, if we use the fact that it does happen.

We could arrive at desired shapes of nanomaterials just by putting it into a process that we know the resulting molecular shapes of. It's like math equations... if $A = B+1$ and C is $B-2$, then what is $A + C$? Here we don't know the actual values of A , B or C , and we need not waste time trying to find out, because we are able to answer the question using the information provided. And that's where we will stop going down this rabbit hole, because we have a limited word count. Let's jump ahead to imagining the future (2030) based on current developments.



Credit: University of Houston

An example of nanoimprint lithography

mold. Photo-nanoimprinting uses a liquid for the resist layer that is UV curable, and a transparent mold, so that the resist and mold layers can be pressed together and then the UV light cures (hardens) the resist layer, making the imprint permanent. Both these methods have an extra step where the etching process is also done on the substrate layer, just as we described for optical nanolithography earlier.

A way to avoid this extra step is to use the direct thermal nanoimprinting method, which basically stamps the substrate or device layer

our gadgets, and to make the manufacturing process orders of magnitude cheaper. While, theoretically, a robot programmed with the skill of a god could take a solid metal bar and etch it away with the precision of a fifth dimension wizard and end up with your favourite phone... such a top down approach is impossible for us mere three dimensional mortals to do.

At the nanoscale, this is no different, and almost all of the wonderful nano creations such as buckyballs, fullerenes, quantum dots, nanotubes and nanowires are all manufactured